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MAT251_TW | Chapter 7: Sequences and Series

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Question **1**

Complete

Not graded

Is the sequence $\{n^2 + n\}$ increasing, decreasing, or neither?

Select one:

- a. Increasing
- b. Decreasing
- c. Neither

The $(n+1)$ th term of the sequence is $(n+1)^2 + (n+1)$, and the nth term of the sequence is $n^2 + n$. Write out $(n+1)^2 + (n+1)$.

$$(n+1)^2 + (n+1) = n^2 + 2n + 1 + n + 1$$
$$= n^2 + 3n + 2.$$

The difference between $n^2 + 3n + 2$ and $n^2 + n$ is

$$n^2 + 3n + 2 - \left(n^2 + n\right) = 2n + 2,$$

which is positive whenever n is positive. Therefore, $(n+1)^2 + (n+1) > n^2 + n$ and the sequence is increasing.

The correct answer is: Increasing

Question **2**

Complete

Not graded

Is the sequence $\left\{ \frac{1}{n^2} \right\}$ increasing, decreasing, or neither?

Select one:

- a. Increasing
- b. Decreasing
- c. Neither

As n gets bigger and bigger, n^2 does, too. In fact, you know that $\left\{ n^2 \right\}$ is a monotonically increasing sequence; in other words, $(n+1)^2 > n^2$. Taking reciprocals, you get $\frac{1}{(n+1)^2} < \frac{1}{n^2}$. Because each term of the sequence is smaller than the one before it, the sequence is decreasing.

The correct answer is: Decreasing

Question **3**

Complete

Not graded

True or false?

The sequence $\left\{ ln\frac{1}{n} \right\}$ converges.

Select one:

True

False

Transform the expression.

$$\ln\frac{1}{n} = \ln\left(n^{-1}\right)$$
$$= -\ln n$$

You know that $\ln x$ is an increasing function; in other words, $\ln(n+1) > \ln n$.

Thus, $-\ln(n+1) < -\ln n$, or

$$\ln\frac{1}{n+1} < \ln\frac{1}{n}.$$

Therefore, the sequence is decreasing.

You also know that $\lim_{x\to\infty} \ln x = \infty$, so as n grows, $\ln\frac{1}{n}$ goes to $-\infty$. Therefore, the sequence is not bounded from below, and so it is divergent.

The correct answer is 'False'.

Question **4**

Complete

Not graded

To which of these values does the sequence $\left\{ e^{(n-n^2)} \right\}$ converge?

Select one:

- a. 1
- b. 1 / *e*
- c. 0
- d. The sequence diverges to −∞

As long as $n \geq -2$, the function $n - n^2$ is strictly decreasing.

Therefore, the sequence $\left\{ e^{\left(n-n^2\right)} \right\}$ is decreasing as well. The terms of the sequence consist of a positive number raised to some power, so they are all positive; therefore, the sequence is bounded below by 0. You have a decreasing sequence that's bounded below, so it converges. If you choose any positive number y, you can find a (negative) power of e that is less than y. Therefore, the sequence is not bounded below by any positive number, so 0 is the greatest lower bound. Thus, the sequence converges to 0.

The correct answer is: 0

Question **5**

Complete

Not graded

Is the sequence $\{(-1)^n n^2\}$ bounded above, bounded below, both, or neither?

Select one:

- a. Bounded above
- b. Bounded below
- c. Both
- d. Neither

There is no number greater than every term of the sequence, and no number less than every term of the sequence. Therefore, the sequence is neither bounded above nor below.

The correct answer is: Neither

Question **6**

Complete

Not graded

To which of the following does the sequence $\left\{ \frac{n}{3^n} \right\}$ converge?

Select one:

- a. 0
- b. 1/3
- c. 1
- d. The sequence diverges to ∞.

The sequence is monotonically decreasing, since $\frac{n+1}{3^{n+1}} < \frac{n}{3^n}$ for all natural numbers $n = 1, 2, \ldots$. To see that this is true, you can look at the ratio of the $n+1$st term to the nth term. If the resulting fraction is less than 1 for all n, the sequence must be monotonically decreasing.

$$\frac{\frac{n+1}{3^{n+1}}}{\frac{n}{3^n}} = \frac{n+1}{n} \cdot \frac{3^n}{3^{n+1}} = \frac{n+1}{3n} = \frac{1}{3} + \frac{1}{3n} < 1$$

for all natural n. Thus, the sequence is monotonically decreasing.

The nth term of this sequence, $\frac{n}{3^n}$, is always positive for $n = 1, 2, \ldots$, so the sequence is bounded below by 0. Combining the facts that the sequence is monotonically decreasing and bounded below by 0, it can be concluded that the sequence has a limit.

Notice that for any natural n, $3^n \geq n^2$, so $\frac{n}{3^n} \leq \frac{n}{n^2}$. Simplifying, $\frac{n}{3^n} \leq \frac{1}{n}$. Thus, for all n, $0 \leq \frac{n}{3^n} \leq \frac{1}{n}$. Use the Squeeze Theorem to find the limit.

$$\lim_{n\to\infty} 0 \leq \lim_{n\to\infty} \frac{n}{3^n} \leq \lim_{n\to\infty} \frac{1}{n}$$

$$0 \leq \lim_{n\to\infty} \frac{n}{3^n} \leq 0$$

Thus $\lim_{n\to\infty} \frac{n}{3^n} = 0$.

The correct answer is: 0

Question **7**

Complete

Not graded

True or false?

The sequence $\left\{ sin\frac{1}{n} \right\}$ converges.

Select one:

True

False

When $n \geq 1$, $\frac{1}{n} \leq 1$. As long as $0 \leq x \leq 1$, the sine function is an increasing function; in other words, if $0 \leq a \leq b \leq 1$, then $\sin(a) \leq \sin(b)$. This is true because the derivative of sine is cosine, and $\cos x \geq 0$ when $0 \leq x \leq 1$. Now

$$0 \leq \frac{1}{n+1} \leq \frac{1}{n} \leq 1,$$

therefore,

$$\sin\left(\frac{1}{n+1}\right) \leq \sin\left(\frac{1}{n}\right).$$

Thus, the sequence is decreasing. Also, $\sin x$ is positive when $0 \leq x \leq 1$, i.e., the sequence is bounded below by 0. Therefore, you have a decreasing sequence which is bounded from below. As a result, the sequence is bounded.

The correct answer is 'True'.

Question **8**

Complete

Not graded

True or false?
The sequence $\{2^{-n}\}$ converges.

Select one:

- True
- False

A sequence is decreasing if each term is less than the one before it. The $(n+1)^{\text{th}}$ term is

$$2^{-(n+1)} = \frac{1}{2^{n+1}},$$

and the nth term is $\frac{1}{2^n}$. If you take the inequality $1 < 2$ and divide both sides by 2^{n+1}, you get the inequality

$$\frac{1}{2^{n+1}} < \frac{1}{2^n}.$$

Therefore, the sequence is decreasing. Because $\frac{1}{2}$ is a positive number, all its powers are positive; therefore, the sequence is bounded below by 0. You have a decreasing sequence that is bounded below; therefore, it converges.

The correct answer is 'True'.

Question **9**

Complete

Not graded

Is the sequence $\{ln\, n\}$ bounded above, bounded below, both, or neither?

Select one:

- a. Bounded above
- b. Bounded below
- c. Both
- d. Neither

The first term of this sequence is ln 1, which equals 0. This sequence is increasing, which means that each term of the sequence is greater than all the ones before it. Therefore, the sequence is bounded from below by any negative number.

The correct answer is: Bounded below

Question **10**

Complete

Not graded

True or false?

The sequence $\left\{ \frac{1}{n^2+1} \right\}$ converges.

Select one:

- True
- False

In order to check if a sequence is increasing or decreasing, you need to check if consecutive terms either always go up or always go down. In other words, you need to determine if either

$$\frac{1}{n^2+1} > \frac{1}{(n+1)^2+1}$$

or

$$\frac{1}{n^2+1} < \frac{1}{(n+1)^2+1}.$$

Expand the denominator on the right.

$$(n+1)^2+1 = n^2+2n+1+1$$
$$= n^2+2n+2;$$

Compare it to the one on the left.

$$n^2+2n+2-\left(n^2+1\right) = 2n+1$$

The difference is greater than 0 whenever n is a positive integer.

Therefore, $(n+1)^2+1 > n+1$;

taking reciprocals produces

$$\frac{1}{n^2+1} < \frac{1}{(n+1)^2+1}.$$

Thus the sequence is decreasing.

Because n^2+1 is always positive, so is its reciprocal; thus, the sequence is bounded from below by 0. You have a decreasing sequence that is bounded from below, so it converges.

The correct answer is 'True'.

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